

The Bacterial Arginine-to-Spermidine Carboxyspermidine Route: A Review

A commissioned review-style synthesis of the aspartate- β -semialdehyde (ASA)/NADPH-dependent spermidine biosynthetic module in bacteria. Search date: 2026-08-11. Citations given as PMIDs.

1. Executive Summary

Most textbook accounts of spermidine biosynthesis describe a single, universal route: putrescine is aminopropylated by spermidine synthase (SpeE), using decarboxylated S-adenosylmethionine (dcSAM) generated by S-adenosylmethionine decarboxylase (AdoMetDC/SpeD). This review concerns a mechanistically and evolutionarily distinct **alternative**: the **carboxyspermidine (ASA-dependent) route**. Here, putrescine is condensed not with dcSAM but with **L-aspartate- β -semialdehyde (ASA)**, in an **NADPH-dependent reductive condensation** catalyzed by **carboxyspermidine dehydrogenase (CASDH / CANSDH)**, and the resulting carboxyspermidine is then decarboxylated by **carboxyspermidine decarboxylase (CASDC / CANSDC / NspC)** to yield spermidine. The upstream diamine, putrescine, is supplied from L-arginine by **SpeA** (arginine decarboxylase $\rightarrow$ agmatine) and **SpeB** (agmatinase $\rightarrow$ putrescine), or by several interchangeable feeder reactions.

The two central discoveries that anchor this review are: (i) the carboxyspermidine route is a *bona fide*, non-SAM pathway that is genetically required for spermidine production, growth, and biofilm formation in organisms such as *Vibrio cholerae*, *Campylobacter jejuni*, *Bacteroides thetaiotaomicron*, and *Agrobacterium tumefaciens* [PMID: 19196710](#); [PMID: 22025614](#); [PMID: 27118128](#); and (ii) it is the **dominant** spermidine pathway in the human gut, stomach, and oral microbiomes, apparently having supplanted the AdoMetDC/SpeE route in those communities [PMID: 22025614](#).

A crucial, non-obvious insight from this investigation is that the **same** two terminal enzymes (CASDH and NspC) produce **either spermidine or its lower homologue sym-norspermidine**, and the product is decided **upstream** by which diamine enters the condensation step: arginine-derived **putrescine** gives spermidine, whereas the aspartate-derived **1,3-diaminopropane (DAP)** branch (via Dat/Ddc) gives sym-norspermidine [PMID: 19196710](#);

[PMID: 9260954](#). This means the "norspermidine" naming of the enzymes (CANS DH/CANS DC) reflects the organism in which they were first purified rather than an intrinsic product specificity. Structurally, CASDH is a three-domain NADPH-dependent reductase related to amino-acid-metabolism dehydrogenases, and NspC is a **group IV (β/α -barrel, fold-type III) PLP decarboxylase**, the same structural family as diaminopimelate decarboxylase, ornithine decarboxylase, and — notably — the upstream enzyme SpeA. The whole module is a striking case of **convergent evolution**: it reaches the same polyamine products as the dcSAM route using nonhomologous, nonanalogous enzymes recruited from central amino-acid and cell-wall metabolism [PMID: 40074085](#); [PMID: 20194510](#).

2. Definition and Biological Boundaries

2.1 What is included

The carboxyspermidine route, as scoped here, comprises four catalytic steps that convert L-arginine to spermidine:

Step	Reaction	Enzyme	Family / activity
1	L-arginine $\rightarrow$ agmatine + CO ₂	SpeA (arginine decarboxylase, ADC)	Biosynthetic ADC; group IV PLP decarboxylase
2	agmatine + H ₂ O $\rightarrow$ putrescine + urea	SpeB (agmatinase)	Arginase/agmatinase (ureohydrolase) superfamily
3	putrescine + ASA + NADPH $\rightarrow$ carboxyspermidine + NADP ⁺	CASDH / CANS DH	Carboxy(nor)spermidine dehydrogenase; three-domain NAD(P) reductase
4	carboxyspermidine $\rightarrow$ spermidine + CO ₂	NspC / CASDC / CANS DC	Carboxy(nor)spermidine decarboxylase; group IV PLP decarboxylase

The two **signature, obligatory** enzymes that define the module — and distinguish it from all other spermidine routes — are **CASDH** (step 3) and **NspC** (step 4). Steps 1 and 2 provide the diamine precursor and are **shared with, and interchangeable across, other polyamine and arginine-catabolic pathways**.

2.2 What should be treated separately (neighboring/confusable processes)

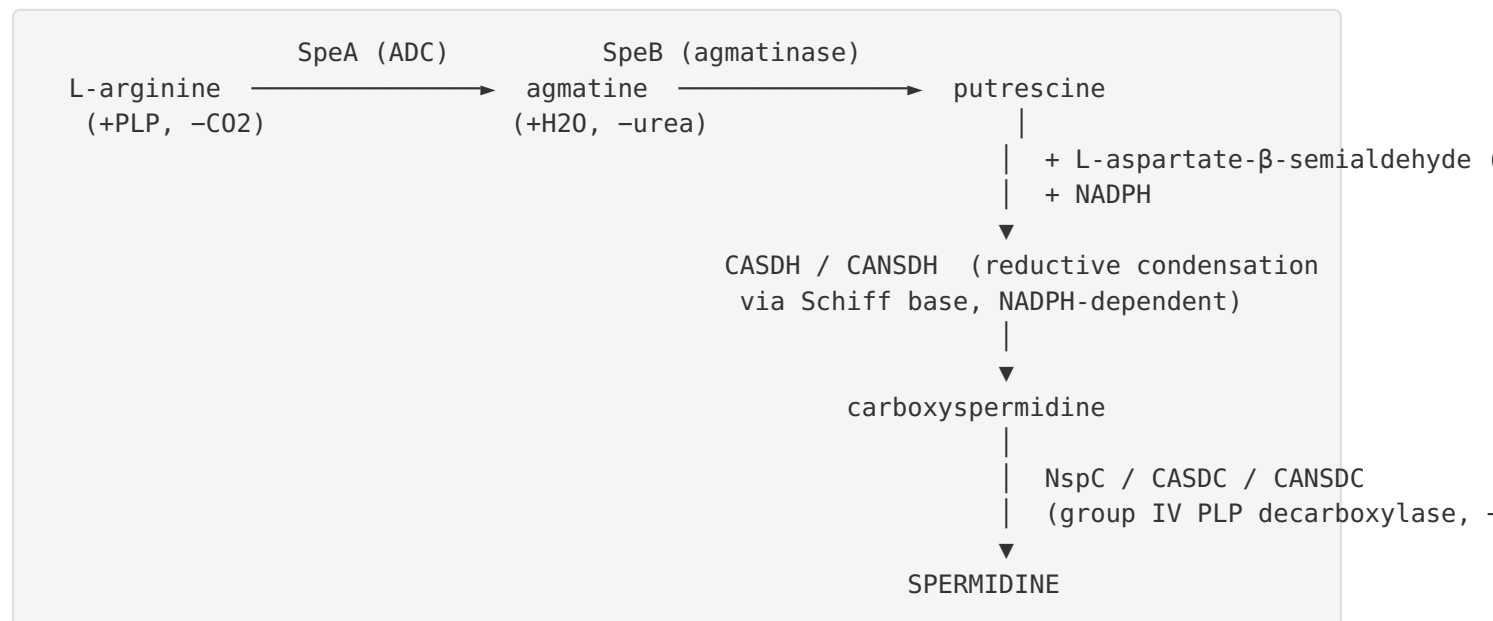
- **The classical dcSAM route (SpeD/SpeE).** This is the *alternative* to the module, not part of it. It draws its aminopropyl unit from SAM/methionine metabolism and uses aminopropyltransferase chemistry (SpeE), whereas the carboxyspermidine route draws its aminopropyl-equivalent from ASA (aspartate/cell-wall metabolism) and uses reductive condensation + decarboxylation. Many microbiome species that use the carboxyspermidine route have degenerate or non-functional SpeE; e.g., *Helicobacter pylori* SpeE lacks substrate-binding residues, does not bind putrescine or dcSAM, and is enzymatically inactive [PMID: 28648602](#).
- **Ornithine decarboxylase (SpeC/ODC) → putrescine.** An alternative *entry* to the putrescine pool that bypasses arginine/SpeA/SpeB. It feeds the same downstream module but is a distinct diamine feeder.
- **Agmatine catabolic operons for energy/pH homeostasis** (e.g., *P. aeruginosa* aguBA and agu2ABCA'), which convert agmatine to putrescine via agmatine deiminase routes and are regulated for biofilm and stationary-phase physiology [PMID: 20149107](#). These overlap chemically at the agmatine→putrescine step but are physiologically separable.
- **Norspermidine sensing/signaling (NspS–MbaA).** In *V. cholerae*, extracellular norspermidine is detected by the periplasmic binding protein NspS and the c-di-GMP-metabolizing enzyme MbaA. This is a *regulatory/signal-transduction* system that is distinct from the *biosynthetic* enzyme NspC, even though both involve norspermidine [PMID: 29045455](#); [PMID: 35302986](#).
- **Homospermidine synthase (HSS) and sym-homospermidine metabolism.** A related "alternative polyamine" branch; HSS evolved *from* CASDH but makes a different product and should be considered a sibling, not part of the core route [PMID: 20194510](#).

2.3 Competing definitions

Nomenclature is the main source of confusion. The dehydrogenase and decarboxylase are variously called **CASDH/CASDC** (carboxyspermidine...) or **CANSDH/CANSDC** (carboxynorspermidine...). Because the identical enzyme pair can make either spermidine or norspermidine depending on the input diamine (§5, §6), these names are best read as denoting an **enzyme family**, not a fixed product. The literature increasingly uses "carboxy(nor)spermidine dehydrogenase/decarboxylase" to acknowledge this.

3. Mechanistic Overview

3.1 Best current model of the sequence of events



Aspartate- β -semialdehyde is supplied by **aspartate- β -semialdehyde dehydrogenase (Asd/ASADH)**, the first branch-point enzyme of the aspartate pathway [PMID: 11724560](#); [PMID: 22683789](#). This couples spermidine synthesis to the metabolic node that also feeds lysine, methionine, threonine/isoleucine, and the cell-wall precursor diaminopimelate.

3.2 Obligatory, conditional, and accessory steps

- **Obligatory (defining) steps:** CASDH (step 3) and NspC (step 4). Genetic ablation of either abolishes the terminal product and produces measurable phenotypes — accumulation of carboxynorspermidine in *V. cholerae* CANSDC mutants, loss of sym-norspermidine/spermidine, 50–60% growth reduction, and severely impaired biofilm formation [PMID: 19196710](#). Deletion of *casdc* in *Bacteroides thetaiotaomicron* depletes spermidine and impairs growth in polyamine-free medium, rescued by complementation [PMID: 27118128](#).
- **Conditional / interchangeable steps:** The diamine-generating steps (arginine $\rightarrow$ agmatine $\rightarrow$ putrescine). These can be provided by SpeA/SpeB, by ornithine decarboxylase (SpeC), by agmatine deiminase routes, or by arginase moonlighting (§5.3). They are conditional on precursor availability and organism.
- **Accessory / upstream shared inputs:** ASA supply (Asd), and — for the norspermidine branch — the Dat/Ddc diamine feeder.

3.3 Molecular assemblies carrying out each step

- **SpeA**: PLP-dependent decarboxylation; structurally a group IV β/α -barrel decarboxylase with a two-subunit shared active site [PMID: 20534592](#).
- **SpeB**: binuclear-metal ureohydrolase of the arginase/agmatinase superfamily.
- **CASDH**: homodimeric three-domain (D1–D3) NADPH reductase (§4.1).
- **NspC**: homodimeric group IV PLP decarboxylase with a shared, dimer-interface active site (§4.2).

4. Major Molecular Players and Active Assemblies

4.1 Carboxyspermidine dehydrogenase (CASDH / CANSDH)

CASDH performs a **transamination-free reductive condensation**: it forms a Schiff base between the amino group of the diamine (putrescine or 1,3-diaminopropane) and the aldehyde of L-aspartate- β -semialdehyde, then reduces it using NADPH to give carboxy(nor)spermidine. The original biochemical characterization in *Vibrio alginolyticus* established the "nicotinamide-nucleotide-linked reduction of the Schiff base," with a strong preference for NADPH over NADH (K_m NADPH $\approx$ 1.5 mM) [PMID: 1955861](#).

Crystal structures of *Helicobacter pylori* CASDH, alone and with NADP, revealed a **three-domain architecture (D1, D2, D3)**; the enzyme **homodimerizes exclusively through D3**, and NADP binds in an interdomain cleft between D1 and D3, triggering domain closure and local rearrangements that assemble the substrate cavity near the nicotinamide moiety [PMID: 36283333](#). This fold is structurally related to lysine/saccharopine-type NAD(P)-dependent amino-acid metabolic enzymes, reinforcing the "recruitment from amino-acid metabolism" theme (§6).

4.2 Carboxyspermidine decarboxylase (NspC / CASDC / CANSDC)

NspC removes the aspartate-derived carboxyl group from carboxyspermidine to give spermidine. Structural and sequence analyses place it firmly in the **PLP-dependent basic amino-acid decarboxylase β/α -barrel class (group IV, fold-type III)** — the same family as meso-diaminopimelate decarboxylase and eukaryotic ornithine decarboxylase [PMID: 20534592](#); an early sequence study of *V. alginolyticus* nspC already noted a PLP-binding region resembling bacterial DAP decarboxylases and eukaryotic ODCs [PMID: 7812450](#).

Structures of ADC (SpeA) and CANSDC (NspC) solved as product complexes (with agmatine and norspermidine, respectively) show that the **active site forms between the β/α -barrel domain of one subunit and the β -barrel of the other**, and that substrate specificity is set by a "specificity helix." In CANSDC, a key acidic residue that binds the distal amino group of other family substrates is replaced by **Leu314**, which instead contacts the aliphatic portion of norspermidine — a compact structural explanation for how a lysine/ornithine-type decarboxylase scaffold was retuned to a polyamine substrate [PMID: 20534592](#).

That **both** the first (SpeA) and last (NspC) decarboxylation steps of the route are carried out by members of the **same group IV PLP fold** is a notable internal symmetry of the pathway.

4.3 Upstream feeders and precursor suppliers

- **SpeA/SpeB (arginine → putrescine)**. Classical, well-characterized in *E. coli*; their genetics and overexpression are long established [PMID: 6392022](#); [PMID: 2440022](#).
- **Asd/ASADH (→ ASA)**. Supplies the aminopropyl donor; catalyzes reductive dephosphorylation of aspartyl- β -phosphate to ASA [PMID: 22683789](#); an essential, human-absent enzyme and antimicrobial target [PMID: 11724560](#).
- **Dat/Ddc (→ 1,3-diaminopropane)**. The alternate diamine feeder that steers the route to norspermidine (§5.4).

5. Evolutionary and Cell-Biological Variation

5.1 Distribution across lineages and dominance in host-associated communities

The carboxyspermidine route is present across many bacterial phyla and is the **dominant spermidine route in the human gut, stomach, and oral microbiomes**, apparently having supplanted the AdoMetDC/spermidine synthase pathway in the gut microbiota [PMID: 22025614](#). Many spermidine-producing species lack functional AdoMetDC/SpeE yet encode CASDH and NspC orthologues; *H. pylori* is a clear example, using the alternative route rather than its inactive SpeE [PMID: 28648602](#). In *C. jejuni*, the CANSDC orthologue makes spermidine (not norspermidine) in vivo, and its deletion compromises growth, rescuable by exogenous polyamines [PMID: 22025614](#).

5.2 Community-level operation (a cell-biological "variation" specific to microbiomes)

In the gut, arginine-to-polyamine flux is a **collective, cross-fed process** rather than a single-species trait. Isotope tracing (^{13}C -arginine) in faecal cultures and rat colon showed putrescine is produced through multiple pathways whose **extracellular intermediates (agmatine, N-carbamoylputrescine) are exchanged between species** [PMID: 30183487](#). Consistent with metabolic division of labor, roughly half of gut Firmicutes are polyamine auxotrophs that nonetheless encode the potABCD polyamine importer [PMID: 22025614](#). Diet/mucin-driven changes in microbial arginine metabolism alter colonic polyamine levels and epithelial barrier function [PMID: 38564708](#).

5.3 Variation in the agmatine → putrescine step

This step is mechanistically the most variable in the route. At least **three non-homologous routes** convert agmatine to putrescine: (i) direct hydrolysis by agmatinase (SpeB); (ii) agmatine deiminase (AIH) → N-carbamoylputrescine → putrescine via N-carbamoylputrescine amidohydrolase (NCPAH); and (iii) AIH → then putrescine transcarbamylase [PMID: 40673658](#). Moreover, enzymes of the closely related **arginase/agmatinase superfamily can moonlight**: in *Thermus thermophilus*, which lacks a canonical agmatinase, an **arginase (TTHA1496)** supplies the physiologically dominant agmatinase activity [PMID: 37001547](#).

5.4 Product variation: spermidine vs. sym-norspermidine is set by the diamine

A key finding of this review is that the **terminal enzymes do not select the product** — the **upstream diamine does**:

Input diamine	Origin	CASDH condensation product	NspC product
Putrescine (4C)	arginine → SpeA/ SpeB	carboxyspermidine	spermidine
1,3-diaminopropane (3C)	aspartate → Dat/ Ddc	carboxynorspermidine	sym-norspermidine

The 1,3-diaminopropane branch is produced by **L-2,4-diaminobutyrate aminotransferase (Dat/DABA AT)** and **DABA decarboxylase (Ddc)**. DABA AT catalyzes the reversible reaction $\text{L-2,4-diaminobutyrate} + \text{2-ketoglutarate} \rightleftharpoons \text{L-glutamate} + \text{L-aspartate-}\beta\text{-semialdehyde}$, tying the diamine directly to the aspartate pathway; ^{14}C -aspartate is incorporated into DAP [PMID: 9260954](#); [PMID: 9514614](#). In *V. cholerae* these activities are present as a **fused DABA aminotransferase/decarboxylase** that supplies DAP to CANSDH, yielding sym-norspermidine independently of arginine [PMID: 19196710](#). Thus the same downstream machinery serves two products, decided by which feeder is active.

5.5 Physiological-state variation: spermine and homospermidine offshoots

The same CASDH/CASDC enzymes can iterate to make **spermine** when spermidine levels are low, constituting the first identified bacterial spermine biosynthetic route and revealing feedback regulation of the branch [PMID: 26682642](#). A sibling enzyme, **homospermidine synthase (HSS)**, evolved from CASDH and diverts the pathway toward sym-homospermidine [PMID: 20194510](#). In *A. tumefaciens*, the essential function of spermidine maps specifically to its **1,3-diaminopropane moiety**, and multiple polyamines bearing that moiety substitute for growth [PMID: 26682642](#).

6. Constraints, Dependencies, and Failure Modes

6.1 Obligatory ordering

The chemistry enforces a strict order: **decarboxylation before condensation before decarboxylation**. 1. SpeA must decarboxylate arginine to make agmatine before SpeB can hydrolyze it. 2. A diamine (putrescine or DAP) and ASA must both be present before CASDH can form and reduce the Schiff base. 3. The carboxyl group introduced by ASA is only present *because* CASDH used ASA; NspC's decarboxylation therefore **must follow** the CASDH step. There is no route to spermidine through this module that skips carboxyspermidine — deletion of NspC/CASDC causes carboxy(nor)spermidine to **accumulate**, direct evidence that the intermediate is obligatory and that no bypass exists [PMID: 19196710](#).

6.2 Substrate-level constraints (what is mutually exclusive)

- **Product choice is mutually exclusive at the diamine step:** a given condensation event uses either putrescine ($\rightarrow$ spermidine) or 1,3-diaminopropane ($\rightarrow$ norspermidine), not both.

- **Aminopropyl-donor source distinguishes the route from dcSAM:** the carboxyspermidine module taps the **aspartate/diaminopimelate** pool (via ASA), whereas the classical route taps **SAM/methionine**. This is a defining, non-interchangeable metabolic dependency [PMID: 11724560](#).

6.3 Dependency on central metabolism (a systemic constraint)

Because ASA sits at the **first branch point of the aspartate pathway** — feeding lysine, methionine, threonine/isoleucine, and cell-wall diaminopimelate — spermidine synthesis via this route is **coupled to cell-wall and essential-amino-acid biosynthesis** [PMID: 11724560](#); [PMID: 18323627](#). This coupling is a plausible reason the route dominates in fast-growing, host-associated bacteria and makes Asd an attractive antimicrobial target (human-absent).

6.4 Failure modes and phenotypes

Perturbation	Consequence	Reference
Δ CANS _{SDH} or Δ CANS _{SDC} (<i>V. cholerae</i>)	Loss of norspermidine/spermidine; 50–60% growth reduction; severely reduced biofilm	PMID: 19196710
Δ CAS _{DC} (<i>casdc</i> , <i>B. thetaiotaomicron</i>)	Spermidine depletion, loss of decarboxylase activity, growth defect; complementation restores	PMID: 27118128
Δ CANS _{SDC} ortholog (<i>C. jejuni</i>)	Growth compromised, rescued by exogenous polyamines	PMID: 22025614
nspC overexpression (<i>V. cholerae</i>)	↑ biofilm/vps, ↓ motility, but no rise in intracellular norspermidine (homeostatic buffering)	PMID: 22239666

6.5 Regulatory decoupling

Notably, NspC enzyme abundance and intracellular norspermidine concentration are **decoupled**: overexpressing nspC increases biofilm and vps expression without raising norspermidine levels, implying feedback that maintains polyamine homeostasis, and indicating that the biosynthetic enzyme (NspC) and the sensor (NspS) provide **independent inputs** into the biofilm regulatory network [PMID: 22239666](#). Downstream, extracellular (imported/environmental) norspermidine — not intracellularly synthesized norspermidine — is the dominant driver of the NspS–MbaA/c-di-GMP biofilm response [PMID: 29045455](#); [PMID: 35302986](#).

7. Controversies and Open Questions

7.1 Enzyme naming vs. biological product. The persistent "norspermidine" labels (CANSDH/CANSDC/NspC) mislead: in most host-associated bacteria the physiological product is **spermidine**, made from putrescine, and only when the DAP feeder is active does the same machinery make norspermidine [PMID: 22025614](#); [PMID: 19196710](#). Whether individual enzymes have measurable diamine preference (kinetic discrimination) versus purely relying on precursor supply is not fully resolved for most organisms.

7.2 Which feeder dominates in vivo, and where. The agmatine→putrescine step has three interchangeable routes plus arginase moonlighting [PMID: 40673658](#); [PMID: 37001547](#). In complex communities the flux is cross-fed among species [PMID: 30183487](#). Quantitative apportionment of flux among these routes in situ remains an open question.

7.3 Organism mixing in mechanistic claims. Much structural work is from *H. pylori* (CASDH) and *V. cholerae*/*V. alginolyticus* (NspC/ADC), while product/physiology data come from *C. jejuni*, *B. thetaiotaomicron*, and *A. tumefaciens*. Generalizing kinetic or regulatory details across these lineages should be done cautiously; this review flags each claim's source organism accordingly.

7.4 Regulation and homeostasis. The mechanism buffering intracellular norspermidine despite NspC overexpression is inferred but not molecularly defined [PMID: 22239666](#). How the spermidine↔spermine and spermidine→homospermidine branch points are regulated under nutrient/stress states is only partly understood [PMID: 26682642](#); [PMID: 20194510](#).

7.5 Evolutionary origin. The route is convergent with the dcSAM pathway, using nonanalogous, nonhomologous enzymes [PMID: 40074085](#). CASDH is an evolutionary progenitor of HSS and is structurally kin to amino-acid dehydrogenases; NspC/SpeA belong to the ancient group IV PLP decarboxylase family shared with lysine biosynthesis [PMID: 20534592](#). The deepest question — whether the ASA-dependent route or the dcSAM route is ancestral in any given lineage, and how often each has been gained/lost/horizontally transferred — is unresolved, though horizontal transfer of the alternative-polyamine machinery is documented [PMID: 20194510](#).

8. Mechanistic Model / Synthesis

The carboxyspermidine route is best understood as a **modular, convergently evolved spermidine factory** built from two dedicated core enzymes bolted onto interchangeable feeders and a central-metabolism precursor:

- **Core (obligatory, defining):** CASDH (NADPH reductive condensation with ASA) → NspC (PLP decarboxylation). These are the fixed identity of the module.
- **Front-end (interchangeable):** any route that fills the diamine pool — SpeA/SpeB from arginine, SpeC from ornithine, agmatine deiminase routes, or arginase moonlighting.
- **Precursor tap (systemic dependency):** ASA from Asd, coupling the route to the aspartate/diaminopimelate/cell-wall node.
- **Product switch (upstream, not terminal):** putrescine → spermidine; 1,3-diaminopropane → sym-norspermidine.
- **Regulatory outputs:** the products (and enzyme NspC) feed into biofilm/c-di-GMP networks, but biosynthesis and sensing are decoupled.

This architecture explains the route's ecological success in host-associated microbiomes: it is metabolically economical (reuses a cell-wall/amino-acid precursor), evolutionarily portable (horizontally transferred, few dedicated genes), and functionally flexible (one enzyme pair, two products, several feeders). It is why organisms with degenerate SpeE nonetheless make abundant spermidine.

9. Limitations and Knowledge Gaps

- **Assay/organism heterogeneity.** Structural, kinetic, genetic, and physiological data come from different bacteria; a single organism with complete structural + flux + regulatory characterization is lacking.
- **In situ flux unknown.** Community cross-feeding [PMID: 30183487](#) means single-species genotype→phenotype inferences may not reflect gut reality.
- **Intrinsic enzyme selectivity underexplored.** Whether CASDH/NspC kinetically discriminate putrescine vs. DAP (beyond precursor supply) is not systematically measured across taxa.

- **Regulation.** The homeostatic buffering mechanism [PMID: 22239666](#) and transcriptional control of the operon are not fully defined.
- **This report is literature-synthesis only** — no new primary data were generated; conclusions rest on the cited studies.

10. Proposed Follow-up Experiments / Actions

1. **Dual-substrate kinetics** of purified CASDH and NspC from a single organism (e.g., *B. thetaiotaomicron* or *C. jejuni*) with both putrescine and 1,3-diaminopropane, to test whether the product switch is purely precursor-driven or partly kinetic.
 2. **Isotope-resolved flux (^{13}C -arginine and $^{13}\text{C}/^{14}\text{C}$ -aspartate)** in defined co-cultures to quantify how much spermidine vs. norspermidine each feeder contributes, and to map cross-feeding of agmatine/N-carbamoylputrescine.
 3. **Structure of the CASDH ternary complex** (enzyme·NADP·carboxyspermidine or a substrate analog) to visualize the reductive-condensation transition state and the diamine-length selectivity determinants, complementing the apo/NADP structures [PMID: 36283333](#).
 4. **Define the norspermidine homeostatic buffer** in *V. cholerae* (efflux, feedback inhibition, or degradation) that decouples NspC level from intracellular product [PMID: 22239666](#).
 5. **Asd/ASADH as an antimicrobial node.** Test whether Asd inhibition selectively starves the carboxyspermidine route in microbiome pathogens without harming SpeD/SpeE users, exploiting the human-absent target [PMID: 11724560](#).
 6. **Phylogenomic reconstruction** across microbiome phyla to date gains/losses/horizontal transfers of CASDH/NspC vs. SpeD/SpeE and identify the best ancestral representatives of each family [PMID: 20194510](#); [PMID: 40074085](#).
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11. Key References

PMID	Title (abbreviated)	Role in this review
19196710	<i>An alternative polyamine biosynthetic pathway...essential for biofilm formation in V. cholerae</i>	Defines the ASA-dependent route; Δ CANS _{SDH} / Δ CANS _{SDC} phenotypes; fused DABA AT/Dc feeder
22025614	<i>Alternative spermidine route...dominant polyamine pathway in human gut microbiota</i>	Dominance in microbiomes; in vivo product is spermidine; auxotrophy/potABCD
36283333	<i>Structural analysis of carboxyspermidine dehydrogenase from H. pylori</i>	CASDH three-domain fold, D3 dimerization, NADP binding
1955861	<i>Carboxynorspermidine synthase in V. alginolyticus</i>	NADPH-dependent Schiff-base reduction mechanism
20534592	<i>β/α-barrel basic amino acid decarboxylases: ADC and CANS_{DC} structures</i>	SpeA and NspC are group IV PLP decarboxylases; Leu314 specificity determinant
7812450	<i>Cloning of nspC from V. alginolyticus</i>	NspC PLP-binding region akin to DAP/ornithine decarboxylases
40673658	<i>Tse8...N-carbamoylputrescine amidohydrolase</i>	Three routes agmatine→putrescine
37001547	<i>Putrescine biosynthesis by arginase in T. thermophilus</i>	Arginase moonlighting as agmatinase
40074085	<i>New routes for spermine biosynthesis</i>	Convergent evolution vs. dcSAM route
20194510	<i>Evolution/horizontal transfer of sym-homospermidine pathway</i>	HSS evolved from CASDH; HGT of alternative-polyamine machinery
11724560	<i>Aspartate semialdehyde dehydrogenase active site</i>	ASA branch-point; human-absent drug target
22683789	<i>ASADH ternary complexes (M. tuberculosis)</i>	ASA-producing reaction
18323627	<i>Second ASADH isoform in V. cholerae</i>	ASA supply / aspartate-pathway coupling
30183487	<i>Luminal putrescine by collective microbiome pathways</i>	Community cross-feeding of intermediates

PMID	Title (abbreviated)	Role in this review
38564708	<i>Mucin/arginine metabolism and colonic barrier</i>	Diet-driven modulation of polyamine output
22239666	<i>NspC overexpression and biofilm without norspermidine rise</i>	Enzyme/product decoupling; biosynthesis vs. sensing
29045455	<i>Norspermidine synthesis vs. signaling in V. cholerae biofilm</i>	Extracellular norspermidine drives NspS–MbaA
9260954	<i>DABA aminotransferase in A. baumannii</i>	Dat reaction links DAP branch to ASA/aspartate
9514614	<i>dat/ddc in H. influenzae</i>	Enzymes making 1,3-diaminopropane
26682642	<i>1,3-diaminopropane moiety essential in A. tumefaciens</i>	Spermine branch; DAP-moiety requirement
28648602	<i>H. pylori does not use spermidine synthase</i>	Degenerate SpeE; reliance on alternative route
27118128	<i>CASDC in B. thetaiotaomicron</i>	Δ casdc depletes spermidine, growth defect
6392022 / 2440022	<i>speA/speB/speC genetics in E. coli</i>	Classical arginine→putrescine feeder
20149107	<i>Agmatine operon in P. aeruginosa</i>	Distinct agmatine catabolic/biofilm system
35302986	<i>c-di-GMP input-output via NspS-MbaA</i>	Norspermidine-responsive signaling distinct from NspC

Prepared as a commissioned review synthesis. Claims are attributed to their source organisms and studies; the reader is cautioned against generalizing kinetic, structural, or regulatory details across bacterial lineages without direct evidence.